Concise Report: Funding Source and Work Completed

ARTI 503 Parallel Architecture and Programming | Assignment 1 | Term 1 2026-2027

Project title	Parallelizing a Compute-Intensive Image Processing Pipeline: Speed-Up, Efficiency and Scalability Analysis of Sequential and Parallel Implementations
Group / leader	Group No. 2 – Moudi Alameel
Selected funding source	AWS Cloud Credits
Instructor	Dr. Rabab Alkhalifa

1. Selected Funding Source

Our group selected AWS cloud to support the project "Parallelizing a Compute-Intensive Image Processing Pipeline: Speed-Up, Efficiency and Scalability Analysis of Sequential and Parallel Implementations". The programme provides AWS cloud credits that can be used for computing and storage services, which our experiments need. We read the eligibility criteria and instructions before applying and checked that the project and group meet them. The instructor was added as a faculty member where the application required it.

2. The Project in Brief

The project compares sequential and parallel implementations of an image-processing pipeline made up of grayscale conversion, Gaussian blur, Sobel edge detection and histogram equalisation. We will parallelise across images and within images, and measure execution time, speed-up, efficiency, scalability and cost across different image sizes and worker counts. The work runs over a 13-week period and the requested budget is about USD 1000 for CPU and GPU cloud instances and storage.

3. Work Completed

- **Group:** Formed the subgroup and confirmed the group in Blackboard.
- **Topic:** Chose a topic that can be implemented, benchmarked and explained: a compute-heavy image-processing pipeline with a clear sequential baseline and several opportunities for parallelisation.
- **Baseline code:** Wrote a sequential baseline in Python that processes a set of images using pixel-level loops, and recorded its running time as the reference (T1) for later speed-up measurements.
- **Related work:** Reviewed public examples of sequential and parallel image-processing projects to guide the design and the evaluation method.
- **Application:** Prepared the application sections: introduction, problem statement, detailed description, suggested solution, budget justification and a 13-week timeline.
- **Peer review:** Exchanged applications with another subgroup, completed the peer evaluation form and revised the application based on the feedback received.
- **Submission:** Submitted the application to AWS on October 4th, 2026, and uploaded screenshots and proof of submission to the group OneDrive folder. Any approval or rejection notice will be uploaded when received.

4. Why this Source was Selected

- **Fit with project needs**: The project is an experimental comparison, so the main need is computing resources: multi-core CPU instances to test scalability with different worker counts, and optionally a GPU instance. AWS provides these through Amazon EC2 instances (multi-core CPU and GPU types), with Amazon S3 or EBS for storing the image datasets and results.

- **Scalability and reproducibility**: AWS gives a consistent, reproducible environment and lets us use more cores than a student laptop has, which is essential for meaningful speed-up and efficiency measurements.

- **Realistic scope**: The project is small and well defined, so the request is modest (about USD 1000) and is suitable for a research credits request.

- **Eligibility and timing**: The eligibility criteria and application requirements are suitable for a student group, and the application can be completed and submitted within the course deadline.

- **Purpose**: The project is academic and the results will be shared with the course instructor, which suits a research-credits programme.

5. Evidence Uploaded

Completed application; this report; screenshots from the application process; proof of submission; the text file naming the evaluating and evaluated subgroups; the completed peer evaluation; and any approval or rejection notice, uploaded when received.

Generative AI was used only as a support tool. The group is responsible for the accuracy, originality and citations of this work.